

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Anti-seize compound, Part Number 3824879, or equivalent
- Cummins® electronic service tool or equivalent

Additional Service Items

- No additional service items required.

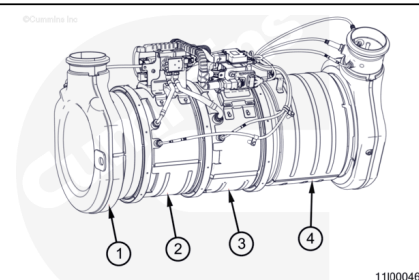
General Information

⚠ WARNING ⚠

The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system, as damage to the catalyst element can result.



Due to the number of varying exhaust aftertreatment applications, this procedure has been written to be generic.

Not all illustrations within this procedure will represent the application being serviced.

The aftertreatment system is composed of four sections. These sections are:

1. The aftertreatment diesel oxidation catalyst (DOC)
2. The aftertreatment diesel particulate filter (DPF)
3. The decomposition reactor tube
4. The selective catalytic reduction (SCR).

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

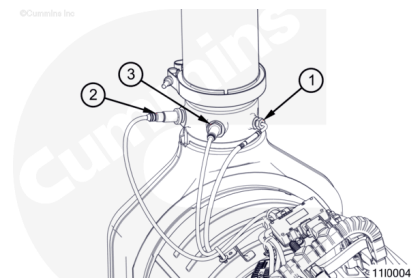
WARNING

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Disconnect the batteries. See equipment manufacturer service information.
- Clean the SCR with compressed air to remove any loose debris.
- If equipped, remove the original equipment manufacturer (OEM) heat shield that attaches to the SCR.
- Draw an orientation reference line across each of the V-band and Torca™ clamps on either end of the aftertreatment SCR canister. This will aid in lining up the clamps and canister to their original orientation during installation.

Remove

Note : Depending upon aftertreatment system installation, the full aftertreatment system may need to be removed from the vehicle prior to disassembly. Refer to Procedure 011-050 in Section 11 (</qs3/pubsys2/xml/en/procedures/493/493-011-050.html>).



If the aftertreatment SCR catalyst is being replaced, complete the following procedures to remove the key parts of the aftertreatment SCR catalyst.

Disconnect the OEM wiring to the aftertreatment interface harness.

Remove the SCR outlet temperature sensor probe (1). Refer to Procedure 019-449 in Section 19.

(</qs3/pubsys2/xml/en/procedures/493/493-019-449.html>)

Remove the aftertreatment particulate matter sensor probe (2). Refer to Procedure 019-478 in Section 19.

(</qs3/pubsys2/xml/en/procedures/493/493-019-478.html>)

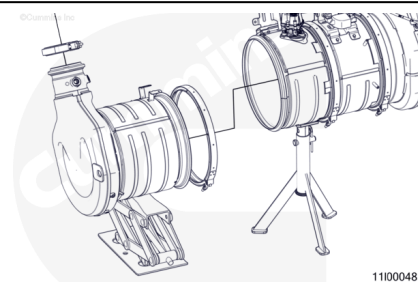
Remove the aftertreatment outlet NOx sensor probe (3). Refer to Procedure 019-451 in Section 19.

(</qs3/pubsys2/xml/en/procedures/493/493-019-451.html>)

Remove any wire ties or P-clips that secure the sensor wires to the SCR.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Support the aftertreatment decomposition tube to prevent it from being freely suspended after the aftertreatment SCR catalyst is removed.

Support the aftertreatment SCR catalyst with a lowering jack.

Remove the clamps on either end of the aftertreatment SCR catalyst.

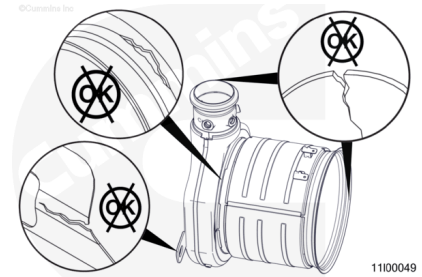
Remove the exhaust hangers that support the aftertreatment SCR catalyst and slowly remove the canister from the vehicle.

Discard the gasket from the V-band joint.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use a metallic object to clean the aftertreatment SCR catalyst. This will scratch the surface of the aftertreatment SCR catalyst which may cause future excessive diesel exhaust fluid (DEF) crystallization buildup.



Note : Refer to Aftertreatment Selective Catalytic Reduction (SCR) Reuse Guidelines, Bulletin5467581, for selective catalytic reduction (SCR) inlet face inspection and cleaning instructions.
(/qs3/pubsys2/xml/en/bulletin/5467581.html)

If buildups are present, carefully scrape the buildup with a non-metallic object to clean the majority of the buildup from the catalyst.

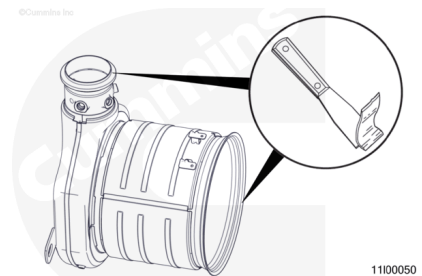
Note : Do not use a mechanical device to remove DEF deposits from the aftertreatment SCR catalyst.

Note : Do not use a pressure washer to remove DEF deposits from the aftertreatment SCR catalyst.

Note : If crystallization buildup was present after installation, complete a stationary regeneration to make sure that any of the remaining DEF crystallization has been removed from the aftertreatment SCR catalyst.

⚠ CAUTION ⚠

Do not grind on the flange surface, as this can damage the flange and cause the connection to leak.



Inspect the exhaust flanges for corrosion or other damage.

Use a putty knife to remove any residual gasket material from the flanges on the aftertreatment SCR catalyst.

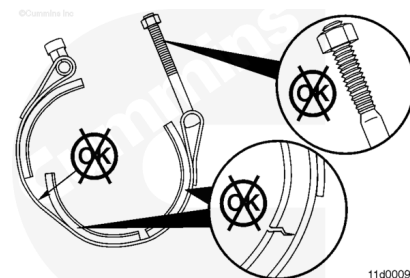
Avoid dropping fragments of gasket material into the aftertreatment SCR catalyst.

Inspect the V-band clamps and mounting straps for signs of over-extension.

The V-band **must not** be bent or damaged.

Inspect the V-band clamp and mounting strap threads for damage.

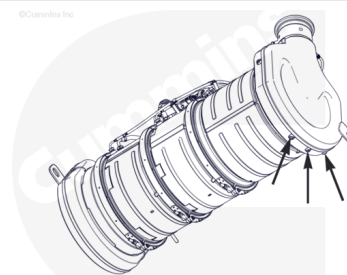
Replace the V-band clamp or mounting strap if damage is found.



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not use an air tool to tighten the V-band clamp. An air tool will damage the threads.

Service locations in California. See Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787 ([/qs3/pubsys2/xml/en/bulletin/6642787.html](https://qs3/pubsys2/xml/en/bulletin/6642787.html)).

Place the aftertreatment SCR catalyst on the vehicle. Use the exhaust hangers to support the canister.

Align the catalyst in the same orientation as marked previously.

If a new horizontal aftertreatment SCR catalyst is being installed, make sure the lowest drain hole is located within ± 10 degrees of the vertical axis and left unplugged.

Apply a coat of anti-seize compound, Part Number 3824879, or equivalent, onto the threads of the V-band clamp. Install a new gasket on the V-band connection of the aftertreatment SCR catalyst and install the opposite of removal. Tighten the V-band and Torca™ clamps.

Torque Value:

SCR Inlet V-band Clamp 20 n•m [177 in-lb]

Note : Do not place the T-bolt section of the V-band clamp near the wiring harness.

Torque Value:

SCR Outlet V-band Clamp 14 n•m [124 in-lb]

Torque Value:

Torca™ Clamp 55 n•m [41 ft-lb]

Tighten the exhaust hangers. See the equipment manufacturer service information.

Install the aftertreatment outlet NOx sensor probe (3). Refer to Procedure 019-451 in Section 19.

(/qs3/pubsys2/xml/en/procedures/493/493-019-451.html)

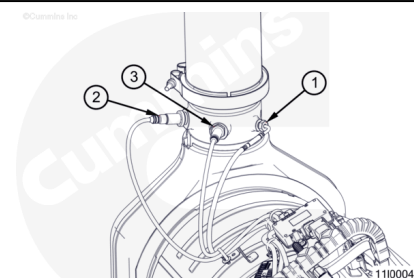
Install the aftertreatment particulate sensor probe (2). Refer to Procedure 019-478 in Section 19.

(/qs3/pubsys2/xml/en/procedures/493/493-019-478.html)

Install the aftertreatment outlet temperature sensor probe. (1)
Refer to Procedure 019-449 in Section 19.

(/qs3/pubsys2/xml/en/procedures/493/493-019-449.html)

Install wire ties or P-clips that secure the sensor wires to the SCR. Make sure all wires are properly secured so that they do **not** come into contact with hot components and areas of wear that can result in damage.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- If equipped, install the OEM heat shield that attaches to the SCR.
- Connect the batteries. See equipment manufacturer service information.
- Start the engine and check the system for leaks.
- Use recommended Cummins® electronic service tool or equivalent to check for active fault codes.
- If buildups were present, complete a stationary regeneration to make sure that any of the remaining DEF crystallization has been removed from the aftertreatment SCR catalyst.
- Continue to monitor for leaks at the exhaust joints.

